

Skills Summary

Reading a Limit Three Ways

Use this reference while completing your worksheet or when studying.

One limit, three kinds of evidence. A graph shows you the shape, a table shows you the trend, and algebra settles the value. Use the first two to predict and to catch mistakes; use the third to prove. Whenever two disagree, the algebra wins — and one of the other two was misread.

Skill 1 — Reading a Limit Off a Graph

Rule: trace the curve toward $x = c$ from each side and read the *height* it heads for. An open circle is a height the graph reaches but does not occupy — it still counts for the limit. A solid dot is $f(c)$ and does **not** count. If the two sides head to different heights, the limit does not exist.

Example: Near $x = 2$ the graph of f follows the line $y = 3x - 1$ on both sides, with an open circle on that line at $x = 2$ and a solid dot at $(2, 1)$. Find $\lim_{x \rightarrow 2} f(x)$.

Step 1. Both sides follow the same line, so the limit is the height that line reaches at 2 — the solid dot is the value $f(2)$, which a limit ignores.

Step 2. On the line, the height at $x = 2$ is $3(2) - 1 = 5$, so both sides head to 5.

$\Rightarrow$ 5 (while $f(2) = 1$, so f is not continuous at 2)

Try it: Near $x = -1$ the graph follows $y = x^2 + 2$ on both sides, with an open circle there and a solid dot at $(-1, 0)$. Find $\lim_{x \rightarrow -1} f(x)$.

check:

Skill 2 — Reading a Limit Off a Table

Rule: inputs must close in from *both* sides. Read the outputs and ask what number they are squeezing toward — not what the last entry says. Two sides settling on different numbers is a jump. A column that keeps growing is an infinite limit. A table can only ever predict: it samples finitely many inputs, so it is evidence, never proof.

Example: A table for $h(x) = \frac{x^2 - 16}{x - 4}$ gives 7.9 at $x = 3.9$ and 8.1 at $x = 4.1$. Find $\lim_{x \rightarrow 4} h(x)$.

Step 1. The two sides are closing in on 8 from below and above, so predict 8 — then prove it, because a table is not a proof.

Step 2. $\frac{x^2 - 16}{x - 4} = \frac{(x - 4)(x + 4)}{x - 4} = x + 4$ for $x \neq 4$, and $4 + 4 = 8$, matching the prediction.
 $\Rightarrow$ 8

Try it: A table for $\frac{x^2 - 100}{x - 10}$ near $x = 10$ gives values just under and just over 20. Find the exact limit.

check:

Skill 3 — Proving It With Algebra

Rule: substitute first. A number is the answer; $\frac{0}{0}$ means rewrite. Three rewrites cover almost everything: factor and cancel, multiply by the conjugate when a square root blocks the cancelling, and split $|x - c|$ by side. After the rewrite, substitute again.

Example: Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+9} - 3}{x}$

Step 1. Substituting gives $\frac{0}{0}$ and nothing factors, so clear the root with the conjugate $\sqrt{x+9} + 3$.

Step 2. $\frac{(x+9) - 9}{x(\sqrt{x+9} + 3)} = \frac{1}{\sqrt{x+9} + 3}$, and at $x = 0$ that is $\frac{1}{6}$.
 $\Rightarrow \frac{1}{6}$

Try it: Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x}$

check: $\frac{0}{0}$